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## R Tutorial 10

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### Instructions:

- Answer all questions.
- Ensure that your findings and results are clearly stated and thoroughly discussed. Please support your arguments using suitable R code with the relevant outputs, interpretations, plots and graphs whenever possible. You should support your argument using appropriate theory that is appropriately referenced.
- The R commands that you use in obtaining your results for all questions must be documented in a R script file. These scripts must be clearly commented. Ensure that any output is clearly stated and interpreted separately from the code as additional comments.
- Include the task name, your name and surname, and your student number in your R script file.
- You MUST label each answer by question number and, where a question has multiple parts, label each part of the question CLEARLY.
- On completion of your assignment, please submit onto RUconnected. If there are any issues uploading onto RUconnected, you may email your submission to: s.izally@ru.ac.za. Please submit your R script file and any other saved data files and plots mentioned in the questions below. Your student number should be included in the name of each file that you submit.
- Each student must complete an individual assignment. You will be assessed based on the quality and/or correctness of the R code, its outputs, and your explanations and interpretations. Acknowledge any help you may have received. Feel free to note any help you may have given to other students in the course.
- This assignment must be submitted by Tuesday, 1 October 2024 by 17:00. Late submissions will be penalized.
- Please note the Rhodes University and the Rhodes University Department of Statistics plagiarism policies.

### Questions:

1. A water bore hole driller will drill a succession of holes in a given area to find a productive well. The probability that he is not successful on any hole (or trial) is 0.8. Let us assume that the probability of a successful hole is independent of any other wells drilled. What is the probability that the third hole drilled will be the second productive well?
2. The employees of a firm that manufactures insulation are being tested for indications of asbestos in their lungs. The firm is requested to send three employees who have positive indications of asbestos on to a medical center for further testing.
  - (a) If 40% of the employees have positive indications of asbestos in their lungs, find the probability that ten employees must be tested in order to find three positives.
  - (b) If each test costs \$20, find the expected value and variance of the total cost of conducting the tests necessary to locate the three positives.
3. Ten percent of the engines manufactured on an assembly line are defective. If engines are randomly selected one at a time and tested,
  - (a) What is the probability that the first non-defective engine will be found on the second trial?
  - (b) What is the probability that the third non-defective engine will be found on the fifth trial?
  - (c) What is the probability that the third non-defective engine will be found on or before the fifth trial?
  - (d) Find the mean and variance of the number of the trial on which the first non-defective engine is found.
  - (e) Find the mean and variance of the number of the trial on which the third non-defective engine is found.
  - (f) Given that the first two engines tested were defective, what is the probability that at least two more engines must be tested before the first non-defective is found?